

torch.nn.attention.bias.CausalBias#

<https://pytorch.org/docs/stable/generated/torch.nn.attention.bias.CausalBias>

Description:

A bias representing causal attention patterns. For an overview of the bias structure, see the CausalVariant enum. This class is used for defining causal (triangular) attention biases. For construing the bias, there exist two factory functions: `causal_upper_left()` and `causal_lower_right()`. This class is a prototype and subject to change.

Function Signature

```
class torch.nn.attention.bias.CausalBias(variant, seq_len_q, seq_len_kv)[source]#
```

Code Examples

```
from torch.nn.attention.bias import causal_lower_right

bsz, num_heads, seqlen_q, seqlen_kv, head_dim = 32, 8, 4, 12, 8

# Create a lower-right causal bias
attn_bias = causal_lower_right(seqlen_q, seqlen_kv)

q = torch.randn(
    bsz, num_heads, seqlen_q, head_dim, device="cuda",
    dtype=torch.float16
)
k = torch.randn(
    bsz, num_heads, seqlen_kv, head_dim, device="cuda",
    dtype=torch.float16
)
v = torch.randn(
    bsz, num_heads, seqlen_kv, head_dim, device="cuda",
    dtype=torch.float16
)

out = F.scaled_dot_product_attention(q, k, v, attn_bias)
```

Notes & Warnings

 **WARNING** This class is a prototype and subject to change.

Detailed Documentation

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